

THE UNIVERSITY OF LAHORE

Course Outline

CS-08519 : Computer Networks Bachelor of Science in Computer Science Winter-2023

Credit Hours	(3+1)
Weekly tuition pattern	2 session lectures (90 min each) 1 Lab (180 min each)
Prerequisites:	None
Course Instructors	Mr. M.Nadeem Akhtar, Mr. M. Shahzad, Ms. Tahseen Fatima and Mr. M Abdullah
Syllabus Designed By	Mr. M. Nadeem Akhtar

1- Course Description:

In today's world, Internet technology is an important part of the communications infrastructure. It is necessary for every student to understand how Internet and its related network technologies works. Especially for Computer Science students, it is critical to understand in detail the working of data communication in computer networks (which are building blocks of internet). It is very important for students to learn how data communication occurs. Moreover, the Internet is an excellent platform for highly distributed and complex computer systems. Consequently, understanding the Internet and its related network technologies is one of major aim of this course.

2- Objectives:

Upon successful completion of the course, a student should be:

1. Able to describe the key terminologies and technologies of computer networks.
2. Able to explain the services and functions provided by each layer in the Internet protocol stack.
3. Capable to identify various internetworking devices and protocols, and their functions in a network .
4. Able to analyze working and performance of key technologies, algorithms and protocols .
5. Capable to build Computer Network on various Topologies.

3- Course Learning Outcomes (CLOs):			
At the end of the course the students will be able to:	Domain	BT Level*	Graduating Attributes (GAs) of PLOs
1.Describe the key terminologies and technologies of computer networks.	C	2	GA1,GA2
2.Explain the services and functions provided by each layer in the Internet protocol stack.	C	2	GA1, GA2
3.Identify various internetworking devices and protocols, and their functions in a network	C	4	GA3
4.Analyze working and performance of key technologies, algorithms and protocols.	C	4	GA3
5.Build Computer Network on various Topologies	P	3	GA4, GA5
* BT= Bloom's Taxonomy, C= Cognitive domain, P= Psychomotor domain, A= Affective domain			
4- Course Contents:			
Introduction and protocols architecture, basic concepts of networking, network topologies, layered architecture, physical layer functionality, data link layer functionality, multiple access techniques, circuit switching and packet switching, LAN technologies, wireless networks, MAC addressing, networking devices, network layer protocols, IPv4 and IPv6, IP addressing, sub netting, CIDR, routing protocols, transport layer protocols, ports and sockets, connection establishment, flow and congestion control, application layer protocols, latest trends in computer networks.			
5- Teaching Methodology:			
Lectures, Written Assignments, Practical labs, Semester Project, Presentations.			
6- Course Assesment:			
Sessional Exam, Home Assignments, Quizzes, Project, Presentations, Final Exam			
7- Text Books:			
1. Data Communication and Computer Networks, 5 th Edition by Behrouz A. Forouzan .			

8- Reference Materials:	
1. Computer Networking: A Top-Down Approach Featuring the Internet, 6th edition by James F. Kurose and Keith W. Ross 2. Computer Networks, 5th Edition by Andrew S. Tanenbaum 3. Data and Computer Communications, 10th Edition by William Stallings	
9- Course Duration:	
This course will be held twice a week of 1.5 hours class duration.	
10- Course style:	
The course will be delivered in a classroom environment.	
11- Additional Course Requirement:	
In addition to the objectives of this course, students are expected to gain skills to model and design the real time organizational computer networks using CISCO Packet Tracer and eNSP simulation tools.	
12- Course Outline:	
The lecturers are supposed to complete the following topics/sub-topics before the mid/final term examination as prescribed in the course outline below:	

Course Delivery Plan		
Week	Lectures	Topics / Sub-Topics
1	1 & 2	Data Communication Fundamentals
		What is Data Communication, Characteristics of Data Communication System, Components of Data Communication System, Data Representation, Data Flow modes Computer Networks Overview Brief Intro to Networks (definition of networks, nodes, links), Computer Network criteria (performance, reliability, security), Physical Structures of Networks (Based on type of connection, Based on Topology)
2	1 & 2	Computer Networks Overview (Cont...) Categories of Networks (LAN,MAN,WAN,PAN,BAN), Internet Protocol Layering: Layering Scenarios and Principles, Why layered architecture? Overview of OSI model. TCP/IP Protocol Suit details, Addressing (Physical, logical, Port, Specific)
3	1 & 2	Transmission Media and Network Connecting Devices

		Physical Layer Data and Signals (Analog as well digital), Parameters of a sine wave (Peak amplitude, frequency/period, phase), Wavelength, Composite signals, Bandwidth, Bit rate, bit length Transmission Impairments (Attenuation, Distortion, Noise), Channel Capacity Definition of channel and channel capacity, Effect of noise on channel capacity, Signal to noise ratio (SNR),
4	1 & 2	Nyquist bandwidth formula (for noiseless channel, Shannon's channel capacity (for noisy channel) Analog and Digital Transmission Overview of different kinds of Conversions (like analog to digital, digital to digital, digital to analog, analog to analog) Digital to Digital Encoding/decoding: Line Coding (Characteristics, Unipolar, Polar, Bipolar,), Block Coding. Assignment 1, Quiz 1
5	1 & 2	Multiplexing Definition and need for multiplexing, Frequency Division Multiplexing (FDM), Wavelength Division Multiplexing (WDM), Time Division Multiplexing (Synchronous, Statistical), Code Division Multiplexing (CDM)
6	1 & 2	Transmission modes: Serial Transmission, Parallel Transmission Circuit Switched and Packet Switched Networks Ethernet LAN (IEEE 802.3) standard, Overview of Wireless LAN
7	1 & 2	Data Link Layer Functions of Data Link Layer, Framing (byte count, character-based framing ,bit-oriented framing), Flow control (stop-and-wait, sliding window, Error detection and correction: Issues related to error detection and correction, Error detection codes (parity check, CRC), Error correction codes (Hamming code), Assignment 2, Quiz 2
8	1 & 2	Data Link Layer (Cont.))MAC protocols Ingredients, Propagation Vs Transmission, Random Multiple Access Protocols (Aloha, slotted Aloha, CSMA and access modes, CSMA/CD)
9	MID-TERM EXAMINATION	
10	1 & 2	Network Layer Connectionless vs connection oriented service, Internet Protocol (IP), IP Addressing (Logical Addressing), Internet Assigned Number Authority (IANA), IPv4 Addresses and Hierarchy, Internet Class-based Addresses, Addressing guidelines
11	1 & 2	Network Layer (Cont ...) Subnet and Sub-netting, Classless Addressing, VLSM, IPv4 Datagram format, Overview of Address Resolution Protocol (ARP), Internet Control Message Protocol (ICMP), Internet Group Message Protocol (IGMP), IPv6 Addressing , Assignment 3, Quiz 3

12	1 & 2	Network Layer (Cont ...) Packet Delivery (Direct, Indirect), Forwarding Techniques (Next hop vs route method, Network specific Vs host specific, Forwarding process, Autonomous systems.
13	1 & 2	Network Layer (Cont ...) Distance vector routing, RIP routing protocol Brief overview OSPF BGP, Assignment 4,
14	1&2	Transport Layer Overview of Transmission Control Protocol, User Datagram Protocol, SCTP, Ports. Quiz 4
15	1&2	Application Layer Protocols Hyper Text Transfer Protocol (HTTP and HTTPS), File Transfer Protocol (FTP), Simple Mail Transfer Protocol (SMTP), POP3 /IMAP4, Telnet, Domain Name Systems (DNS), DHCP.
16		Final Term course overview
FINAL EXAMINATION		

13- Assessment Criteria:

ASSESSMENT CRITERIA			
No.	Assessment	Percentage	
1.	Quizes	10%	
2.	Assignments	10%	
3.	Mid Term Exam	20%	
4.	Final Term Exam	40%	
5.	Lab Work (Lab Tasks + Assignments)	20 %	
Total		100%	

14- Attendance Requirements:

You are expected to attend all lectures, seminars, tutorials, and lab sessions or any other classroom activity announced. Where you fail to attend classes, you cannot expect the lecturer to brief you on what you have missed. You are responsible for your attendance, not the academic staff. Attendance at tutorials and lab sessions will be strictly monitored, and failure to attend will be taken into account.

Note: Minimum of 75% Attendance in lectures/lab sessions/seminars (if any) are required for a student to sit in the Final-Term examination.

Etiquette: Please keep all cell phones turned OFF during class. If your activities during class are deemed disruptive, you will be asked to leave.

15- Submission and Collection of Assignment:

All assignments should be handed in at the beginning of the class sessions when they are

due. All assignments may be handed back during scheduled classes.

16- General Information:

Students are required to be familiar with the university code conduct, and to abide by its terms and conditions.

16.1 Copying of Copyright Material by Student

A condition of acceptance as a student is the obligation to abide by the University's policy on the copying of copyright material. This obligation covers photocopying of any material using the University's photocopying machines, and the recording off air, and making subsequent copies, of radio or television broadcasts, and photocopying textbooks. Students who flagrantly disregard University policy and copyright requirements will be liable to disciplinary action under the Code of Conduct.

16.2 Academic Misconduct

Please refer to the Code of Conduct for definitions and penalties for Academic Misconduct, plagiarism, collusion, and other specific acts of academic dishonesty. Academic honesty is crucial to a student's credibility and self-esteem, and ultimately reflects the values and morals of the University as a whole. A student may work together with one or a group of students discussing assignment content, identifying relevant references, and debating issues relevant to the subject. Academic investigation is not limited to the views and opinions of one individual, but is built by forming opinion based on past and present work in the field. It is legitimate and appropriate to synthesize the work of others, provided that such work is clearly and accurately referenced. Plagiarism occurs when the work (including such things as text, figures, ideas, or conceptual structure, whether verbatim or not) created by another person or persons is used and presented as one's own creation, unless the source of each quotation or piece of borrowed material is acknowledged with an appropriate citation. Encouraging or assisting another person to commit plagiarism is a form of improper collusion and may attract the same penalties. To prevent Academic Misconduct occurring, students are expected to familiarize themselves with the University policy, the Subject Outline statements, and specific assignment guidelines. Students should also seek advice from Subject Leaders on acceptable academic conduct.

16.3 Guidelines to Avoid Plagiarism

Whenever you copy more than a few words from any source, you must acknowledge that source by putting the quote in quotation marks and providing the name of the author. Full details must be provided in your bibliography. If you copy a diagram, statistical table, map, etc., you must acknowledge the source. The recommended way is to show this under the diagram. If you quote any statistics in your text, the source should be acknowledged. Again full details must be provided in your bibliography. Whenever you use the ideas of any other author you should acknowledge those, using the APA (American Psychological Association) style of referencing.

Students are encouraged to co-operate, but collusion is a form of cheating. Students may use any sources (acknowledged of course) other than the assignments of fellow students. Unless your Subject Leader informs you otherwise, the following guideline should be

used: Students may work together in obtaining references, discussing the content of the references and discussing the assignment, but when they write, they must write alone

16.4 Referencing For Written Work

Referencing is necessary to acknowledge others' ideas, avoid plagiarism, and allow readers to access those others' ideas. Referencing should:

1. Acknowledge others' ideas
2. Allow readers to find the source
3. Be consistent in format and
4. Acknowledge the source of the referencing format

To attain these qualities, the school recommends use of either the Harvard or American Psychological Association style of referencing, both of which use the author/date.

16.5 Referencing Standards

APA style referencing

17- Approval:

Checked by: <i>Dr. M. Nadeem Ashraf</i> (Cluster Head)	Approved by: <i>Dr. Adnan Khalid</i> (HOD CS & IT)
---	---